Stock Market Analytics and Time Series Forecasting

Professional Report Based on the Four-Page Power BI Dashboard

This report interprets the four supplied dashboard pages: Market Overview, Portfolio & Risk Analysis, NIFTY 50 Forecasting & Model Comparison, and Regression & Correlation Analysis. Values are reported as displayed in the dashboard; the underlying calculations and source data were not independently audited.

1. Executive Summary

The dashboard combines historical market monitoring, portfolio risk measurement, short-horizon index forecasting, and stock-versus-market sensitivity analysis. For the selected HDFC Bank view, the dashboard reports a latest close of ₹735.00, total return of 23.39%, CAGR of 3.24%, annual volatility of 25.66%, Sharpe ratio of -0.11, and maximum drawdown of -40.47%. These values show that positive displayed total return can coexist with substantial volatility and a deep historical drawdown.

For the equal-weight portfolio of HDFC Bank, ICICI Bank, Infosys, Reliance Industries and TCS, each with a 20% weight, the dashboard reports a ₹10,00,000 portfolio value, historical 95% VaR of ₹17,075.35, historical 95% CVaR of ₹28,568.86, parametric 95% VaR of ₹19,975.45, 30-day Monte Carlo VaR of ₹92,649.97, and a simulated probability of loss of 42.34%. ARIMA(0,1,0) is selected over Prophet based on the displayed holdout metrics. HDFC Bank's reported beta of 1.09 and R-squared of 57.58% indicate positive, above-market estimated sensitivity and a meaningful—but incomplete—linear relationship with NIFTY 50 daily returns.

2. Page 1 — Market Overview

The Market Overview page displays HDFC Bank's adjusted-close history and daily-return fluctuations.

Metric	Dashboard value
Latest close	₹735.00
Total return	23.39%
CAGR	3.24%
Annual volatility	25.66%
Sharpe ratio	-0.11
Maximum drawdown	-40.47%

The adjusted-close chart gives long-term price context across approximately 2020–2026. The daily-return chart highlights short-term variation and occasional large positive or negative moves. Annual volatility of 25.66% indicates meaningful return variability. Maximum drawdown of -40.47% represents a substantial peak-to-trough decline at some point in the displayed history. The negative Sharpe ratio indicates that the measured excess return relative to the risk-free rate was negative under the dashboard's calculation method, despite the positive displayed total return.

Validation note: total return and CAGR measure performance differently. Before external use, confirm the date range, return definitions, risk-free-rate assumption and annualisation conventions behind these KPIs.

3. Page 2 — Portfolio & Risk Analysis

The portfolio page compares the equal-weight portfolio with NIFTY 50, displays the five equal 20% allocations, and summarizes historical, parametric and Monte Carlo risk.

Risk indicator	Dashboard value
Portfolio value	₹10,00,000.00
Historical VaR (95%)	₹17,075.35

Historical CVaR (95%)	₹28,568.86
Parametric VaR (95%)	₹19,975.45
Monte Carlo VaR (30-day)	₹92,649.97
Probability of loss	42.34%

Historical VaR estimates a downside threshold from observed returns, while historical CVaR estimates the average loss in the tail beyond that threshold. Parametric VaR uses a distributional model and may differ from historical VaR. The 30-day Monte Carlo VaR is larger because it measures a 30-trading-day simulated horizon rather than a single day. The 42.34% probability of loss means that 42.34% of generated terminal values finished below the initial ₹10 lakh within the simulation.

The displayed 5th-percentile terminal value is approximately ₹9,07,350. Subtracting this from ₹10,00,000 gives a loss of approximately ₹92,650, consistent with the reported 30-day Monte Carlo VaR. The simulation depends on historical mean returns and covariance relationships and the model's multivariate-normal return assumption. It is a scenario analysis, not a guarantee. VaR figures should not be compared without noting their different time horizons and methodologies.

4. Page 3 — NIFTY 50 Forecasting & Model Comparison

The forecasting page compares ARIMA and Prophet using a 60-day holdout evaluation and presents a 30-business-day ARIMA forecast with a 95% confidence interval. The dashboard identifies ARIMA(0,1,0) as the best model.

Metric	ARIMA(0,1,0)	Prophet
MAPE	2.26%	7.14%
RMSE (NIFTY 50 points)	626.68	1,735.80
MAE (as shown in error chart)	544.30	1,705.80

ARIMA has lower displayed MAPE, MAE and RMSE than Prophet on the selected holdout period, indicating smaller errors for these metrics. MAPE expresses average absolute percentage error; MAE is average absolute error in index points; RMSE penalizes larger errors more heavily. The 95% confidence interval communicates forecast uncertainty, not a guaranteed range for future index values.

Model selection should be limited to the tested evaluation setup: lower holdout error does not guarantee better future performance if market conditions change. The PDF does not provide sufficiently readable exact forecast values or interval endpoints for reliable transcription, so these are not inferred in this report.

5. Page 4 — Regression & Correlation Analysis

For the selected HDFC Bank regression against NIFTY 50 daily returns, the dashboard reports:

Metric	Dashboard value
Beta	1.09
Annualised alpha	-7.03%
R-squared	57.58%
Residual volatility	16.71%
Daily alpha	-0.0279%

A beta of 1.09 indicates positive estimated sensitivity to NIFTY 50 movements: in the fitted relationship, a 1% index return is associated with about a 1.09% HDFC Bank return on average. This is a statistical relationship, not a rule that holds each day. R-squared of 57.58% means the fitted market-return model explains about 57.58% of the observed variation in HDFC Bank returns in the sample; the remainder is not explained by this one-factor relationship. Negative alpha values indicate a negative estimated intercept/excess-performance measure under the dashboard's conventions. Residual volatility of 16.71% represents the reported variability not explained by the fitted market relationship.

6. Correlation Findings

The displayed daily-return correlation matrix shows positive correlations across all listed assets. The strongest stock-to-stock relationship shown is Infosys–TCS at 0.73. HDFC Bank and ICICI Bank each correlate 0.76 with NIFTY 50; Reliance Industries correlates 0.71, Infosys 0.57 and TCS 0.55 with NIFTY 50. HDFC Bank and ICICI Bank correlate at 0.63.

These positive correlations suggest that the assets often moved in the same direction historically, although the strength varies by pair. Diversification benefits may be limited when several holdings respond similarly to broad market movements. Correlations are historical estimates and may change.

7. Integrated Interpretation

Together, the pages connect performance tracking, downside risk, forecast evaluation and market sensitivity. The market page shows that positive cumulative return does not eliminate volatility or drawdown risk. The portfolio page quantifies downside exposure and demonstrates that a positive simulated average can coexist with a meaningful probability of a negative 30-day outcome. The forecasting page favours ARIMA over Prophet on the displayed holdout metrics. The regression page indicates positive market sensitivity for HDFC Bank, while the correlation matrix shows that the selected stocks are not independent of one another.

8. Recommendations

- Document the data frequency, forecast target, training period, holdout window and evaluation formulas so model comparisons are reproducible.
- Use rolling or walk-forward validation and periodically reassess whether ARIMA remains the best-performing model.
- Display VaR and CVaR with confidence level, currency and time horizon; clearly separate one-day risk from 30-day Monte Carlo risk.
- Supplement Monte Carlo results with stress scenarios or historical shock analysis because normal-return simulations may underrepresent extreme events.
- Check consistent definitions for total return, CAGR, Sharpe ratio, alpha and annualisation before comparing metrics.
- Treat beta, alpha and correlations as historical estimates, not permanent characteristics or standalone investment signals.
- Use the dashboard as an analytical monitoring tool rather than a standalone investment recommendation.

9. Conclusion

The Power BI dashboard demonstrates an integrated workflow for Indian equity time-series analysis, including performance monitoring, portfolio risk measurement, NIFTY 50 forecasting, and market relationship analysis. ARIMA(0,1,0) performs better than Prophet on the displayed holdout metrics, while the portfolio and regression pages highlight meaningful downside risk and market exposure. The dashboard is most useful for transparent monitoring and comparison when metric definitions, risk horizons, forecast uncertainty and model limitations are clearly communicated.